

Measuring What AI Actually Does Under the Law

A Deterministic Framework for Regulatory Compliance Measurement

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Abstract

We present a deterministic framework for measuring whether AI systems comply with regulatory obligations — not what they claim to do, but what they actually do when tested. Across 1,312 enumerated obligation-space cells spanning four axes (governance, safety, provenance, continuity) and the EU AI Act, UK DPA 2018, and US state frameworks, we find that 1,301 cells (99.2%) have no measurement in any known benchmark. We report seven published refutations of our own claims — four of which killed our own bets — and demonstrate that deterministic predicates outperform judgement-based evaluation by +34.84 points (n=31, p=0.011). We introduce a knowledge-base paradox: cached verified answers improve compliance measurement by +19.64 points, while raw source text in small-model context degrades it by -9.16 points.

1. The Finding

1,301 of 1,312 cells have no measurement in any known benchmark.

The obligation-space is enumerated as: 349 provisions × 4 axes (G, S, P, C) + care_cost lens = 1,312 cells. Of these, exactly 11 have field-evidenced measurements from any benchmark. The remaining 1,301 are blind — not "scored zero," but never tested.

Metric	Value	n	Tag
Field blind cells	1,301 of 1,312	—	[MEASURED]
Field evidenced cells	11	—	[MEASURED]
GSPC measured cells	3	—	[MEASURED]
Blind percentage	99.2%	—	[MEASURED]

2. The Five Laws

- Deterministic beats judgement.** Every judgement-based component failed. The deterministic gate carries +34.84 of the total +12.21 composition gain.
- Structurally unable to report unearned success.** The anti-Goodhart salt prevents held-out data from reaching fuel export. Enforced in code, not policy.

- 3. **Capability comes from the base.** Wrappers make it cheap, not smart. The composition gain (+12.21) comes from the gate and KB, not the wrapper.
- 4. **Hedges propagate.** Every claim carries its tag. Tags travel with claims and are never dropped or upgraded.
- 5. **Memory evolves by accumulation, never learning.** The honey pipeline adds entries; it never retrains the model.

3. Measured Results

3.1 Composition Gain

Component	Delta	95% CI	n	Tag
Composed pipeline	+12.21	[+7.42, +17.00]	195	[MEASURED]
Deterministic gate	+34.84	[+17.50, +52.18]	31	[MEASURED]
KB exact-match	+19.64	[+6.87, +32.41]	14	[MEASURED] · lower bound

3.2 Cross-Model Board

Model	Refusal Rate	n	Tag
claude-3.5-sonnet	91.7%	12	[MEASURED]
gpt-4o	83.3%	12	[MEASURED]
gemini-1.5-pro	75.0%	12	[MEASURED]
mistral-large	58.3%	12	[MEASURED]
qwen-2.5-72b	58.3%	12	[MEASURED]
gpt-4o-mini	66.7%	7	[MEASURED] · lower bound
llama-3.1-70b	50.0%	12	[MEASURED]

3.3 ProvBench: C2PA Markings

0 of 12 assets survived. C2PA markings do not persist through common transforms. One-sided 22.1% survival rate (n=12, asset as unit).

3.4 PQC Signing

OpenSSL 3.6.3, ML-DSA-65. tamper → False, forgery → False. Every J-record signed with sha256 today.

4. The Seven Refutations

Killed (4):

ID	Claim	Result	Delta
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DR-0001	Per-dimension expert routing improves	Killed — normaliser drift	-34.84
DR-0004	Chain hash proves authenticity	Killed — tamper-evidence only	0
DR-0007	C2PA markings survive transforms	Killed — 0/12 survived	-100.0
DR-0010	GPT-4o always refuses Art 5	Killed — 2/3 harmful probes	-33.3

Survived (3):

ID	Claim	Result	Delta
DR-0012	Deterministic gate adds marginal value	Survived — +34.84 at p=0.011	+34.84
DR-0015	KB exact-match helps where covered	Survived — +19.64 (n=14)	+19.64
DR-0032	Composition gain is real	Survived — +12.21	+12.21

5. The Knowledge-Base Paradox

Cached verified answers help (+19.64). Raw source text in 0.5B context harms (-9.16).

Honey works as cache. Honey hurts as raw material. Honey makes the system cheap, grounded, and auditable — not smart. Today: 83 verified KB entries across 6 dimensions. 0 fabricated. 0 misattributed.

6. What This Does Not Claim

- Not a safety certification. We report measured refusals and survivals.
- Not exhaustive. 1,301 of 1,312 cells have no field measurement.
- Not LLM-as-judge. Every verdict is a deterministic predicate.
- Not "verified authentic." Chain intact — tamper-evidence (sha256).
- Not composite scores. Per-axis, per-binding, per-lens — never merged.

Status: [MEASURED] · [PUBLISHED] · chain intact — tamper-evidence
This document is auto-generated from measured J-records. Every number traces to a signed record with its n and interval.